

A Literal Smoothing Counterexample to a Non-Elliptic FIO Schatten Question

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Status

Candidate counterexample, likely valid for the literal class-membership wording. This packet concerns arXiv:2408.06833 by Cardona, Delgado, and Ruzhansky [1]. It does not claim to settle the corrected exact-order variant discussed in Section 1.

Source Problem

The source paper asks the following in its Main Results section, PDF page 5.

sharp when T is elliptic, and leaving open the following question:

Open problem 1.1. Prove (or disprove) that if $T \in I_{1,0}^\mu(X; \Lambda)$, is a non-elliptic Fourier integral operator that belongs to the Schatten class $S_r(L^2(X))$, then $\mu < -n/r$, when $r \in (0, 1) \cup ((1, \infty) \setminus \mathbb{Z})$.

To the best of our knowledge, under the conditions in Open problem 1.1, one has the validity of the order inequality $\mu \leq -n/r$, see Lemma 4.13.

Transcribed, the open problem asks whether, for

$$r \in (0, 1) \cup ((1, \infty) \setminus \mathbb{Z}),$$

every non-elliptic Fourier integral operator

$$T \in I_{1,0}^\mu(X; \Lambda)$$

that belongs to the Schatten class $S_r(L^2(X))$ must satisfy

$$\mu < -n/r.$$

Definitions and Convention

Let X be a compact smooth manifold of positive dimension n , equipped with a smooth positive density so that $L^2(X)$ is fixed. We write Δ for the diagonal canonical relation. The standard smoothing operators satisfy

$$\Psi^{-\infty}(X) \subseteq \Psi_{1,0}^\mu(X) = I_{1,0}^\mu(X; \Delta)$$

for every real μ ; equivalently, their kernels are C^∞ and their finite-order principal symbols vanish. This is the ordinary class-membership interpretation of $T \in I^\mu$, as opposed to saying that T has exact order μ .

Proof Intuition

Schatten membership cannot force a strict upper bound on a merely declared order class, because smoothing operators belong to all finite-order classes at once. A rank-one smoothing operator is as compact as possible: it has only one nonzero singular value, so it lies in every Schatten class. But if we view the same operator as an element of $I_{1,0}^\mu$ for an arbitrary large μ , it is still non-elliptic because its principal symbol at that order is zero. Thus the order parameter in the literal hypothesis is not an intrinsic order of the operator.

Theorem 1. *Let X be a compact smooth manifold of positive dimension n . For every $r > 0$ and every $\mu \geq -n/r$, there exists a nonzero non-elliptic operator $T \in I_{1,0}^\mu(X; \Delta)$ such that*

$$T \in S_r(L^2(X)).$$

Consequently the literal statement of the source open problem is false.

Proof. Choose nonzero functions $u, v \in C^\infty(X)$. Define

$$Tf = \langle f, v \rangle_{L^2(X)} u, \quad f \in L^2(X).$$

The Schwartz kernel is

$$K_T(x, y) = u(x) \overline{v(y)},$$

which is C^∞ on $X \times X$. Hence T is smoothing:

$$T \in \Psi^{-\infty}(X).$$

By the usual nesting of pseudodifferential symbol classes, $\Psi^{-\infty}(X) \subseteq \Psi_{1,0}^\mu(X)$ for every real μ . Since pseudodifferential operators are Fourier integral operators associated with the diagonal canonical relation, this gives

$$T \in I_{1,0}^\mu(X; \Delta)$$

for every real μ .

The operator T has rank one. More explicitly,

$$T = u \otimes v$$

in the Hilbert-space rank-one notation, so its singular values are

$$s_1(T) = \|u\|_2 \|v\|_2, \quad s_j(T) = 0 \quad (j \geq 2).$$

Therefore

$$\sum_{j \geq 1} s_j(T)^r = (\|u\|_2 \|v\|_2)^r < \infty$$

for every $r > 0$. Thus $T \in S_r(L^2(X))$, including the quasi-Banach range $0 < r < 1$.

Finally, T is not elliptic as an element of any finite-order class $I_{1,0}^\mu(X; \Delta)$. Its finite-order principal symbol is zero because the operator is smoothing. In particular it is non-elliptic. If $\mu \geq -n/r$, the conclusion $\mu < -n/r$ fails. Taking $\mu = 0$ gives a counterexample for every $r > 0$ and every $n \geq 1$. $\square$

Verification Notes

The proof is symbolic and does not rely on computation. The included script `code/rank_one_schatten_check.py` verifies the finite-dimensional rank-one singular-value pattern used in the argument. The command used was:

```
conda run --no-capture-output -n sandbox python code/rank_one_schatten_check.py
```

The script is only a sanity check. The proof of smoothing/FIO membership is the standard kernel-class argument above.

1 Limitations and Exact-Order Variant

This counterexample uses the difference between membership in a filtered class and exact order. It does not address the strengthened statement:

If T is non-elliptic but has exact order μ , or has a nonzero principal symbol at order μ , and $T \in S_r(L^2(X))$, must $\mu < -n/r$?

That is the natural repaired version of the problem. The source paper itself uses exact-order language in the proofs of the trace-class and integer-index results; for instance it argues with T “having order μ ” and with a nonzero principal symbol. The present packet therefore should be read as a literal counterexample and a repair note: the stated open problem needs an exact-order or nonzero-principal-symbol hypothesis to avoid smoothing operators.

Novelty Check

The local run indexes were searched for the arXiv id, the paper title, and the phrases

```
non-elliptic Fourier integral,  mu < -n/r,  
smoothing Fourier integral Schatten,  rank-one smoothing.
```

No prior run packet or attempt for this target was found.

A bounded web search on 2026-06-28 for 2408.06833, the title, and the exact open-problem phrase surfaced the source arXiv page but no later answer or discussion of this smoothing counterexample. Novelty confidence is therefore moderate: the example is elementary and may be considered an implicit convention issue by specialists, but no indexed duplicate was found.

Human Review Recommendation

Check the source authors’ intended convention for $T \in I^\mu$. Under the ordinary class-membership convention, the counterexample is decisive. If the authors intended exact order, the packet should be logged as a wording repair and not as a disproof of that strengthened problem.

References

- [1] D. Cardona, J. Delgado, and M. Ruzhansky, *Nuclearity, Schatten-von Neumann classes, distribution of eigenvalues and L^p - L^q -boundedness of Fourier integral operators on compact manifolds*, arXiv:2408.06833, 2024.
- [2] L. Hörmander, *The Analysis of Linear Partial Differential Operators III: Pseudo-Differential Operators*, Springer, 1985.